

PEYTON RAMSEY

(860) 515-6245 | ramseypeyton@gmail.com | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

EDUCATION

University of New Hampshire	Durham, NH
MS in Business Analytics (MSBA), GPA: 3.63	August 2025 – May 2026
BS in Business Administration: Focus in Information Systems & Business Analytics	August 2021 – May 2025
<i>NCAA Division I Cross Country & Track, Dean's List, America East Commissioner's Honor Roll</i>	

EXPERIENCE

Graduate Research Assistant, Data Science & AI	August 2025 – May 2026
<i>University of New Hampshire</i>	<i>Durham, NH</i>
- Developed course content and hands-on labs for AI Tools and Apps for Business, teaching practical ML deployment, LLM workflows, Git, Docker, and responsible AI to MSBA students. - Modernized 12+ modules in Predictive Analytics and Big Data, migrating course content from R Markdown to Quarto.	
Data Engineer Intern View TPA Compliance Report	June 2025 – August 2025
<i>Nassau Financial Group (Insurance / Financial Services)</i>	<i>Hartford, CT</i>
- Built and deployed a self-serve R Shiny compliance review platform that prioritized higher-risk cases across 120K+ insurance policies from 7 AWS data streams, reducing manual review time by about 60%. - Engineered ETL pipelines and a configurable weighted scoring engine across 8 insurance activity types; partnered with compliance and stakeholders to deliver a production tool used by 30 analysts and presented to 50+ stakeholders.	
Business Intelligence & Data Intern	June 2024 – August 2024
<i>Sprague Operating Resources (Energy / Natural Gas)</i>	<i>Portsmouth, NH</i>
- Wrote SQL to join and aggregate trade and pricing tables into star-schema models in Power BI for portfolio pricing analysis. - Embedded Python scripts in Power BI to unblock data-refresh issues and complete a risk management dashboard.	

SELECTED DATA SCIENCE PROJECTS

Airbnb Booking Prediction & Market Analysis View Report	March 2025 - May 2026
- Started with a San Francisco booking-conversion study, comparing Random Forest and LightGBM on 12,947 listings using stratified 5-fold CV; the best model reached 0.917 ROC-AUC and highlighted discounts and bedrooms as key drivers. - Expanded into a 374K U.S. classifier and a 93K-row Paris (Popincourt) panel with leakage-safe tabular and NLP features, achieving 0.8668 AUC on Kaggle and 0.966 holdout AUC on the Paris model. - Identified a ~6x booking-rate gap between commercial and casual hosts that price and location could not close.	
Financial News Retrieval and Sentiment Analysis (RAG) Live Demo	January 2026 - Present
- Built a retrieval-augmented generation system over 500+ financial news articles, combining dense retrieval and FinBERT sentiment analysis to answer analyst-style questions about sectors, tickers, and macro themes. - Designed a 5-stage article filtering and embedding pipeline that reduces 500+ raw articles to ~50 high-quality documents. - A/B tested retrieval parameters and index sizes to balance precision and latency, then deployed the system via Docker on Hugging Face Spaces with a live interface for interactive analyst queries.	
Loan Default Prediction and Drift Monitoring View Report	December 2025
- Built a loan default prediction pipeline on 148K applications with 51 engineered features; selected a Random Forest over XGBoost and a PyTorch baseline after comparative evaluation, achieving 0.92 ROC-AUC and 0.86 F1. - Deployed the model via FastAPI and Docker with MLflow experiment tracking, exposing single/batch prediction endpoints. - Implemented real-time PSI, KS-test, and Chi-squared drift monitoring to trigger model-risk review.	

TECHNICAL SKILLS

Programming: Python (Pandas, NumPy, Polars, PyTorch, scikit-learn), SQL, R (tidyverse, dplyr, ggplot2)

Machine Learning & Statistics: supervised & unsupervised learning, classification & regression, feature engineering, model evaluation & cross-validation, tree-based & boosting methods (Random Forests, LightGBM, CatBoost), SHAP, NLP, deep learning (LSTM autoencoders), clustering (k-means, DBSCAN), time series analysis (ARIMA, VECM), A/B testing

MLOps & Tools: FastAPI / Flask, MLflow, Docker, Git/GitHub, Hugging Face, R Shiny, Power BI, AWS